Method 2: the helium atom energy is

$$E = 2\left(\frac{p^2}{2m} - \frac{2ke^2}{r}\right) + \frac{ke^2}{2r} = \frac{\hbar^2}{mr^2} - \frac{7ke^2}{2r}$$

This can be rewritten as

$$E = \frac{\hbar^2}{m}\left(\frac{1}{r} - \frac{7ke^2m}{4\hbar^2}\right)^2 - \frac{49k^2e^4m}{16\hbar^2} .$$

The first term on the right vanishes at minimum energy. The ground energy and radius

$$E_0 = -\frac{49k^2e^4m}{16\hbar^2} , \quad r_0 = \frac{4\hbar^2}{7ke^2m}$$

are respectively as in （7），（ 6）.

2.（ I） The particle crosses the lead plate upward. Its charge is positive.

（II） Velocity v is normal to the field; Lorentz force is in the paper. Magnetism changes momentum direction, not magnitude. Neglecting gas drag, in the steady field the path is a circular arc of constant radius. The equation of motion is

$$qBv = \left|\frac{\triangle p}{\triangle t}\right| = \frac{p\triangle\phi}{\triangle t} = \frac{pv}{r}, \qquad (1)$$

Here q is charge, v speed and p momentum; $\triangle\phi$ is the angle turned in time $\triangle t$, and r is the curvature radius. Hence

$$p = qBr . \qquad (2)$$

$q = e$; let p_d, p_u be entering and exiting momenta. In （1），substitute the data to obtain

$$p_d = 63.0\ \text{MeV/c} , \quad p_u = 22.8\ \text{MeV/c} . \qquad (3)$$

For $pc \gg mc^2$, use special relativity. From

$$p = \frac{mv}{\sqrt{1-(v/c)^2}} . \qquad (4)$$

we obtain

$$v = \frac{c}{\sqrt{1+(mc/p)^2}} . \qquad (5)$$

Let v_d, v_u be entering and exiting speeds. Insert （2） and $m = 0.51\ \text{MeV}/c^2$ into （3） to obtain

$$v_d \approx c , \quad v_u \approx c . \qquad (6)$$

The mean speed through lead is $\bar{v} = (1/2)(v_d + v_u) \approx c$ $\triangle t$ is the transit